

column keyword is not present in the constructor, the columns will be arranged in alphabetical order. Now, if we execute this cell, the result will be a table like this:

	year	team	wins	draws	losses
0	2010	FCBarcelona	30	6	2
1	2011	FCBarcelona	28	7	3
2	2012	FCBarcelona	32	4	2
3	2010	RMadrid	29	5	4
4	2011	RMadrid	32	4	2
5	2012	RMadrid	26	7	5
6	2010	ValenciaCF	21	8	9
7	2011	ValenciaCF	17	10	11
8	2012	ValenciaCF	19	8	11

where each entry in the dictionary is a column. The index of each row is created automatically taking the position of its elements inside the entry lists, starting from 0. Although it is very easy to create DataFrames from scratch, most of the time what we will need to do is import chunks of data into a DataFrame structure, and we will see how to do this in later examples.

Apart from DataFrame data structure creation, Panda offers a lot of functions to manipulate them. Among other things, it offers us functions for aggregation, manipulation, and transformation of the data. In the following sections, we will introduce some of these functions.

Open Government Data Analysis Example Using Pandas

To illustrate how we can use Pandas in a simple real problem, we will start doing some basic analysis of government data. For the sake of transparency, data produced by government entities must be open, meaning that they can be freely used, reused, and distributed by anyone. An example of this is the Eurostat, which is the home of European Commission data. Eurostat's main role is to process and publish comparable statistical information at the European level. The data in Eurostat are provided by each member state and it is free to reuse them, for both noncommercial and commercial purposes (with some minor exceptions).

Since the amount of data in the Eurostat database is huge, in our first study we are only going to focus on data relative to indicators of educational funding by the member states. Thus, the first thing to do is to retrieve such data from Eurostat. Since open data have to be delivered in a plain text format, CSV (or any other delimiter-separated value) formats are commonly used to store tabular data. In a delimiter-separated value file, each line is a data record and each record consists of one or more fields, separated by the delimiter character (usually a comma). Therefore, the data we will use can be found already processed at book's Github repository as `educ_figdp_1_Data.csv` file. Of course, it can also be downloaded as unprocessed tabular data from the Eurostat database site¹³ following the path: